

Strategic Academic and Professional Development Plan

Goal 1 Applied Machine Learning Foundations

Objective

Strengthen practical understanding of core machine learning concepts. This includes regression and classification, as well as clustering and model evaluation.

Outcomes

Complete five end to end machine learning projects. Cover different problem types. Each project must include data preprocessing and model training. Include evaluation metrics and interpretation of results.

Action Steps

Build projects using banking or customer behavior datasets. Use linear regression and decision trees. Apply k-means clustering. Document everything in Jupyter notebooks.

Context

This supports growth as a machine learning engineer. It turns theory into practice.

Timeline

Eight to ten weeks.

Goal 2: Advanced Data Analysis proficiency

Objective

Master cleaning and visualizing real world datasets. Use Python and modern analytical tools.

Outcomes

Finish four full exploratory data analysis reports. Include statistical summaries and visualizations. Identify patterns and correlations. Find deep insights.

Action Steps

Use pandas and NumPy. Work with business or social data. Maintain a consistent structure across all datasets.

Context

EDA is the baseline for data science. Reliable models require clean data.

Timeline

Four to six weeks.

Goal 3 SQL and Querying Competence

Objective

Write efficient SQL queries for data extraction.

Outcomes

Solve thirty SQL problems on a recognized platform. Move from basic queries to joins. Master subqueries and aggregations.

Action Steps

Practice two problems daily. Review mistakes and rewrite code. Focus on logic.

Context

SQL is a non negotiable skill for technical interviews. It is the industry standard.

Timeline

Four weeks.

Goal 4 Professional Portfolio Development

Objective

Create a public portfolio. Show off data analysis and machine learning work.

Outcomes

Host six documented projects on a version control platform. Include clear problem statements.

Show methodology and results.

Action Steps

Refine internship and course projects. Improve documentation. Ensure the code is easy to reproduce.

Context

A strong portfolio creates visibility. It proves skills to recruiters.

Timeline

Six to eight weeks.

Goal 5 Professional Career Readiness

Objective

Prepare for internships or entry level roles.

Outcomes

Complete one recognized data science course. Participate in a technical interview simulation.

Action Steps

Finish remaining course sessions. Review concepts every week. Practice explaining technical decisions.

Context

This bridges the gap between school and the corporate world. It secures the future.

Timeline

Five to six weeks.

Execution Roadmap

Milestone	Target Period	Core Focus
Phase 1	Weeks 1 to 4	SQL and EDA

Phase 2	Weeks 5 to 10	ML Foundations
Phase 3	Weeks 11 to 14	Portfolio and Readiness

Would you like me to create a weekly study schedule based on these goals?